Last Name: _____ **First Name:** _____

SUNet ID: _____@stanford.edu

Midterm Solutions — October 23, 2026

Closed-book exam. No notes, no laptop, no phone, no internet, no AI assistance, no anything. Just your brain and a pen!

Duration: 1 hour 30 minutes **Total number of points: 100**

Instructions

Questions are grouped into logical sections to ease your thought process. There are two types of questions:

- **Multiple-choice**: One correct choice per question. Please (circle) the correct answer.
- **Free-form**: Give concise answers and address each requested result and explanation.

In all cases, there is no penalty for wrong answers.

I. Transformers (25 points)

1. (1 point) Two tokenizers encode the same teddy bear story without losing information. Tokenizer A produces twice as many tokens as B. For this story, with the same model context limit, which conclusion follows?

   A. A doubles the dimension of each token embedding.

   B. A needs twice as many entries in its token vocabulary.

   C. A uses twice as many positions in the context window.

   D. A can fit twice as much of the story in a single context.

2. (1 point) The token `bear` refers to a teddy bear in one sentence and means "tolerate" in another. Its token ID is unchanged. Which statement distinguishes an embedding lookup from a contextual Transformer representation?

   A. Both must differ because the surrounding words differ.

   B. The lookup is the same, but the contextual representation can differ.

   C. The lookup changes, and attention restores a shared representation.

   D. Both must be the same because the token ID is unchanged.

3. (1 point) All input tokens of a vanilla RNN are known during training. Why can its hidden states still not all be computed independently in parallel?

   A. The full sequence must be generated before its inputs are known.

   B. Each position uses different recurrent weights.

   C. Each hidden state depends on the preceding hidden state.

   D. Each token requires an embedding update.

4. (2 points) An encoder–decoder translates a sentence about a teddy bear. The source has 5 tokens and the decoder prefix has 3 tokens. For one cross-attention head, where do Q and K, V come from, and what is the shape of $QK^\top$?

   A. Q, K, V: decoder, shape 3×3.

   B. Q, K, V: encoder, shape 5×5.

   C. Q: decoder, K, V: encoder, shape 3×5.

   D. Q: encoder, K, V: decoder, shape 5×3.

5. (2 points) In "A cute teddy bear is reading", a head gives high attention weight from `bear` to `reading`. What does this imply?

   A. The value vector for `reading` has a large weight in `bear`'s output.

   B. The weight is the probability of generating `reading` next.

   C. The output at `bear` is replaced by `reading`'s input embedding.

   D. The attention weight from `reading` to `bear` is equally high.

6. (2 points) Two attention heads have identical Q, K, V projections, dimensions, masks, and input. With no dropout, what follows before concatenation and output projection?

   A. Each output is scaled by the number of heads.

   B. Softmax assigns them distinct attention patterns.

   C. Their output vectors must be orthogonal.

   D. The heads produce identical weighted sums of the values.

7. (2 points) One token's feature vector and the parameters of a standard per-token LayerNorm are fixed. Other tokens and batch examples change. What happens to that token's LayerNorm output?

   A. It changes: the variance uses all batch examples.

   B. It changes: the mean uses all sequence positions.

   C. It is unchanged: its features and the LayerNorm parameters are fixed.

   D. It changes: normalization depends on how many tokens the batch contains.

8. (1 point) Compared with $y = F(x)$, what does a residual connection $y = x + F(x)$ add to backpropagation?

   A. A direct gradient path that bypasses F.

   B. A guarantee that gradients never vanish across layers.

   C. A constraint that makes the derivative of F positive.

   D. A normalization step that fixes the gradient norm.

9. (3+4 points) **Representing word order.** Teddy and Plushie are teddy bears. Consider "Teddy follows Plushie" and "Plushie follows Teddy", with one token per word. A Transformer encoder uses full self-attention, no position information, and shared per-token operations. Ignore dropout. Each sentence is represented by the average of its final token vectors.

   (i) Must the two sentence representations be identical? Explain how reordering the input tokens affects the final token vectors and their average.

   (ii) Now use causal self-attention with no positional embeddings. Each token can attend only to itself and earlier tokens. Must the averages still match? Compare the prefixes at `follows` and explain whether the permutation argument from (i) still applies.

   (i) - **Token vectors (1 point):** The final token vectors are permuted in the same way as the input tokens.

   - **Reason (1 point):** Full attention with no position information and shared per-token operations preserves this property through every layer.

   - **Average (1 point):** Yes. Reordering the token vectors leaves their average unchanged.

   (ii) - **Teddy prefix (1 point):** At `follows`, "Teddy follows Plushie" has the visible prefix "Teddy follows".

   - **Plushie prefix (1 point):** At `follows`, "Plushie follows Teddy" has the visible prefix "Plushie follows".

   - **Token vectors (1 point):** The permutation argument no longer applies: reordering tokens changes which tokens can interact through causal attention.

   - **Averages (1 point):** They need not match, since the final token vectors need not be permutations of each other.

10. (3+3 points) **What label smoothing asks the model to do.** A classifier labels teddy bear reviews as positive (1), neutral (2), or negative (3). For one review, the true class is 1 and the smoothed target is $q = (0.90, 0.05, 0.05)$. The loss is $L(p) = -\sum_{j=1}^{3} q_j \log p_j$, where p is the model's predicted distribution over the three classes.

    (i) With the target q fixed, someone claims that the loss is minimized when the model's predicted distribution p equals $(1, 0, 0)$. Is this correct? Give the minimizing p and justify your answer using $L(p)$.

    (ii) Now consider next-word prediction after "The teddy bear is". Why can label smoothing be useful when each training example supplies only one next word? Discuss valid alternative words, model confidence, and generalization.

    (i) - **Verdict (1 point):** No.

    - **Minimum (1 point):** $p = q$.

    - **Reason (1 point):** $p_2 = p_3 = 0$ while $q_2, q_3 > 0$, so the $\log 0$ terms prevent a finite loss.

    (ii) - **Alternatives (1 point):** Several next words can be valid, such as `soft` or `brown`.

    - **Confidence (1 point):** Discourages putting all probability on the observed word.

    - **Generalization (1 point):** Can reduce overfitting to the observed continuations.

II. Large Language Models (25 points)

1. (1 point) In "The teddy bear sat by the bank of the river," `river` resolves the meaning of `bank`. Comparing a bidirectional encoder and a causal decoder, which can use `river` in its final hidden state at `bank`?

   A. Both can use the right-hand word if they have enough layers.

   B. Only the causal decoder can use the right-hand word.

   C. Neither can use the right-hand word until it is generated again.

   D. Only the bidirectional encoder can use the right-hand word.

2. (2 points) A sparse MoE layer has 8 experts and routes each token to 2 experts. Compared with evaluating all 8 experts for every token, what does this design change?

   A. It chooses one fixed pair of experts for all tokens in every sequence.

   B. It evaluates all 8 experts, then discards 6 of their output vectors.

   C. It evaluates 2 experts per token while retaining 8 experts' parameters.

   D. It retains only 2 experts' parameters for the entire input sequence.

3. (2 points) An attention layer has 8 query heads. Each key and value head has dimension 64. Replace MHA with GQA using 2 key/value groups, keeping all 8 query heads. How many key *plus* value scalars are produced per token?

   A. 256: 2 groups times 64 dimensions times 2 for keys and values.

   B. 128: 2 groups times 64 dimensions, counting one vector per group.

   C. 1,024: 8 query heads times 64 dimensions times 2 for keys and values.

   D. 512: 8 query heads times 64 dimensions, counting one vector per head.

4. (2 points) RoPE rotates a given query at position m and a given key at position n. If both position indices increase by the same offset, their dot-product score:

   A. Changes in proportion to the offset, regardless of token content.

   B. Changes because RoPE makes the score depend on the sum $m + n$.

   C. Moves toward zero because both absolute positions are larger.

   D. Stays the same because the relative displacement is unchanged.

5. (1 point) After "The teddy bear is", the next-token probabilities for (`soft`, `brown`, `small`, `worn`) are $(0.60, 0.25, 0.10, 0.05)$. With top-p sampling at $p = 0.80$, which tokens are retained before renormalization?

   A. `soft`, `brown`, `small`.

   B. `soft`, `brown`.

   C. `soft` only.

   D. `soft`, `brown`, `small`, `worn`.

6. (2 points) For fixed logits with a unique maximum, temperature increases from $T = 1$ to $T = 2$. No top-k or top-p filtering is used. Which statement is correct in exact arithmetic?

   A. Greedy and sampling both become uniform over all vocabulary tokens.

   B. Greedy decoding changes its token, but sampling probabilities stay fixed.

   C. Greedy keeps its token, and sampling becomes more concentrated on it.

   D. Greedy keeps its token, while sampling probabilities become flatter.

7. (1 point) Guided decoding enforces a JSON schema for a teddy bear inventory, with an integer `count` field. The count has the right type but is factually wrong. What does this show?

   A. Format constraints do not establish factual correctness.

   B. The schema must enumerate more tokens to verify the number of bears.

   C. The schema constraint must have failed during token generation.

   D. Using greedy decoding would make the integer factually correct.

8. (1 point) For a teddy bear counting problem, self-consistency selects the most common final answer among sampled reasoning paths. Why can it still select a wrong answer?

   A. It gives longer reasoning paths more voting weight.

   B. It votes over token frequencies.

   C. The paths can share the same mistake.

   D. Sampling makes each answer equally reliable.

9. (3+2+2 points) **Local choices and sequence probability.** A model generates a two-token label for a teddy bear. Let $a =$ `soft`, $b =$ `brown`, $u =$ `bear`, and $v =$ `toy`. First-token probabilities are $P(a) = 0.6$, $P(b) = 0.4$. The only possible second tokens are u, v, with

$$P(u \mid a) = 0.5, \quad P(v \mid a) = 0.5, \qquad P(u \mid b) = 0.9, \quad P(v \mid b) = 0.1.$$

Ties are broken in favor of u. There is no length penalty.

(i) Give the greedy sequence and its probability. Compute all four complete-sequence probabilities to identify the most probable sequence.

(ii) Which first-token prefixes does beam search of width 2 retain? Use them to explain whether it recovers the most probable sequence here.

(iii) In a real language task, would finding the most probable sequence guarantee a factually correct answer? Explain.

(i) - **Greedy (1 point):** au (`soft bear`), with probability $0.6 \cdot 0.5 = 0.30$.

- **Sequence probabilities (1 point):** au, av, bu, bv have probabilities 0.30, $0.30, 0.36, 0.04$.

- **Most probable (1 point):** bu (`brown bear`) is the most probable sequence, at 0.36.

(ii) - **Prefixes (1 point):** Width 2 retains both a and b at the first step.

- **Conclusion (1 point):** Yes. Comparing their extensions recovers bu.

(iii) - **Distinction (1 point):** No. Model likelihood does not establish factual correctness.

- **Explanation (1 point):** A model error can make a false answer most probable.

10. (4+2 points) **How far can local attention reach?** A causal Transformer reads `My teddy bear wears a bow` (one token per word) using two attention layers. In each layer, position t can attend to positions $t, t-1, t-2$ of the preceding layer, when they exist. All other operations act on each position separately. There are no global tokens or other links between positions.

    (i) Give the earliest input position that can influence position 6 after both layers. Show a path, justify the maximum backward reach, and state whether input position 1 can influence it.

    (ii) Give one architecture change that would let input position 1 influence position 6, and state its computation or communication-path trade-off.

    (i) - **Position (1 point):** Position 2 (`teddy`).

    - **Path (1 point):** Output 6 $\leftarrow$ layer-1 4 $\leftarrow$ input 2.

    - **Bound (1 point):** Each layer reaches back at most 2 positions, so the maximum reach is $2 + 2 = 4$ positions across two layers.

    - **Exclusion (1 point):** Position 1 (`My`) cannot influence output 6.

    (ii) - **Change (1 point):** Third layer ($6 \leftarrow 4 \leftarrow 2 \leftarrow 1$), wider window, or suitable global link.

    - **Trade-off (1 point):** Extra layer computation, or more attention work for a wider/global link with a shorter path.

III. LLM training (25 points)

1. (1 point) A pretrained model predicts held-out teddy bear stories well but often continues an instruction instead of following it. Which explanation best fits this observation?

   A. The model needs a new vocabulary before instruction tuning can work.

   B. Low next-token loss implies that instruction examples were mislabeled.

   C. Predicting text well does not ensure the desired assistant behavior.

   D. Pretraining optimizes a sentence-level instruction success reward.

2. (1 point) An engineer puts three solved teddy bear counting examples in a prompt, then runs a frozen model. What distinguishes this from supervised finetuning on those examples?

   A. The examples update only the output projection during inference.

   B. The examples condition inference without updating model weights.

   C. The examples become permanent training targets after generation.

   D. The examples train the embedding layer while freezing attention.

3. (2 points) A causal language model is trained with teacher forcing on "The teddy bear smiles". Treat each word as one token. After "The teddy", its most likely next token is `toy`, not `bear`. Which text does it use as context when computing the training loss for `smiles`?

   A. "The teddy bear", using the words from the training sentence.

   B. "The teddy toy", using the model's most likely prediction.

   C. "The teddy", stopping before the incorrectly predicted word.

   D. "The teddy bear smiles", using the complete training sentence.

4. (2 points) A Bradley–Terry reward model uses $P(y_w \succ y_l \mid x) = \sigma(r(x, y_w) - r(x, y_l))$, where y_w and y_l are the preferred and rejected responses. Add the same constant $c(x)$ to every response score for a prompt. What changes?

   A. Pairwise preference probabilities stay the same.

   B. Preference probabilities move closer to one half.

   C. A negative constant reverses the ranking.

   D. Pairwise preferences stay fixed only if the original scores are positive.

5. (1 point) An RLHF-trained model writes teddy bear stories. Why does its objective discourage large deviations from the starting model?

   A. To force exact copies of the starting model's stories.

   B. To reduce reward hacking and training instability.

   C. To make a separate reward model unnecessary.

   D. To ensure the reward model perfectly matches human preferences.

6. (2 points) Human raters write the best teddy bear story for each prompt. A team finetunes a model on these prompt–story pairs with next-token loss. Which method is this?

   A. DPO, because the training stories reflect human preferences.

   B. SFT, using the written stories as targets without a pairwise preference loss.

   C. Reward modeling, because each story is an example of a good response.

   D. RLHF, because the training responses were provided by humans.

7. (2 points) A frozen teacher generates fresh teddy bear stories before each update of a smaller student model. The student is trained to predict the tokens in those stories. Is this on-policy training for the student?

   A. No, because the teacher supplies the responses used for training.

   B. No, because token-level supervision can only be used off-policy.

   C. Yes, because the training targets are generated instead of human-written.

   D. Yes, because fresh stories are generated before every student update.

8. (1 point) Two models use different tokenizers on the same teddy bear stories. Why is bits per byte useful for comparing their losses?

   A. It converts both models to a shared vocabulary before scoring.

   B. It uses a common unit of text length despite different token boundaries.

   C. It averages loss across vocabulary entries instead of text positions.

   D. It gives each story equal weight regardless of length.

9. (4+3 points) **Adapting with LoRA.** A model is adapted to write teddy bear stories. One linear layer has frozen base weights $W_0 \in \mathbb{R}^{1024\times512}$. LoRA uses $W = W_0 + BA$ with rank parameter $r = 2$. Ignore biases and any scaling constant.

   (i) Give the shapes of A and B, calculate the number of trainable parameters, and identify which matrices are updated or frozen.

   (ii) A team increases r from 2 to 16, keeping the base model and training data fixed. Explain how this changes the flexibility of the weight update and the memory needed to train the adapter. Does it guarantee better teddy bear stories on unseen prompts? Justify your answer.

(i) - **Shape of A (1 point):** $A \in \mathbb{R}^{2\times512}$.

- **Shape of B (1 point):** $B \in \mathbb{R}^{1024\times2}$.

- **Count (1 point):** $2 \cdot 512 + 1024 \cdot 2 = 3{,}072$ trainable parameters.

- **Updates (1 point):** Only A, B are updated, while W_0 stays frozen.

(ii) - **Flexibility (1 point):** The update BA can have higher rank, allowing a wider range of weight changes.

- **Memory (1 point):** More trainable parameters require more memory for weights, gradients, and optimizer state.

- **Generalization (1 point):** No guarantee: the larger adapter may overfit, or rank 2 may already be sufficient.

10. (3+2+1 points) **Comparing DPO policies.** For a prompt asking for a teddy bear story, response y_w is preferred to y_l. The frozen reference assigns probabilities 0.20 and 0.10. Policy A equals the reference. Policy B instead assigns 0.10 and 0.025. Remaining probability mass belongs to other responses. With $\beta = 1$, use

$$m = \log \frac{\pi(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \log \frac{\pi(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}, \qquad L = -\log \sigma(m),$$

where m is the margin and σ is the sigmoid function.

(i) Compute each policy's margin and use the margins to justify which policy has lower DPO loss.

(ii) Which policy assigns higher probability to the preferred response? Compare the policies' preferred/rejected odds to explain why your answer is consistent with (i).

(iii) Is y_w necessarily Policy B's most probable response? Briefly justify by considering the other responses.

(i) - **Margin A (1 point):** $m_A = 0$.

- **Margin B (1 point):** $m_B = \log(0.5) - \log(0.25) = \log 2$.

- **DPO loss (1 point):** B has lower loss because its margin is larger.

(ii) - **Probability (1 point):** Policy A assigns more probability to y_w: $0.20 > 0.10$.

- **Relative odds (1 point):** B's preferred/rejected odds are 4, versus A's 2. DPO rewards higher relative odds even if the preferred response's absolute probability falls.

(iii) - **Other responses (1 point):** No. A response outside this pair could have probability above 0.10.

IV. Reinforcement learning with LLMs (25 points)

1. (1 point) A program returns reward 1 for a correct answer to a teddy bear counting problem and 0 otherwise. The test is not differentiable. Can a policy-gradient method still train the generator?

   A. No: replace the test with a differentiable reward model first.

   B. Yes: the reward weights a gradient of the response log-probability.

   C. Yes: use the derivative of the binary test with respect to the tokens.

   D. No: a policy gradient requires a correct target token at every step.

2. (1 point) In token-by-token generation viewed as reinforcement learning, which pair correctly identifies the state and action at a decoding step?

   A. State: prompt and generated prefix, action: the next token.

   B. State: final reward, action: the response's score.

   C. State: prompt alone, action: the hidden state at the current position.

   D. State: model parameters, action: a gradient update to those parameters.

3. (1 point) Two teddy bear stories for the same prompt receive rewards 8 and 6. Subtracting a baseline of 7 gives advantages $+1$ and -1. Why use these advantages in the policy-gradient update?

   A. To make both stories equally likely by centering their training signal.

   B. To identify which individual tokens caused each story's final reward.

   C. To reduce gradient variance by comparing rewards with a baseline.

   D. To bound the change in each story's probability after an update.

4. (1 point) A PPO value model receives a prompt and an unfinished response prefix. What quantity is it intended to estimate?

   A. The human preference label between this prefix and another prefix.

   B. The exact reward of the best possible completion of this prefix.

   C. The probability of the next token under the frozen reference policy.

   D. The expected future return from this prefix under the policy.

5. (2 points) PPO reuses a batch of teddy bear stories for several updates. The stories were sampled before the policy changed. Why does its objective include a ratio of current-policy to sampling-policy probabilities?

   A. To compute advantages without training a separate value model.

   B. To account for the policy change while reusing the sampled actions.

   C. To keep the policy close to the initial model even after many batches.

   D. To turn each story's reward into a separate reward for each token.

6. (2 points) At a fixed prefix, on-policy distillation minimizes reverse KL, $D_{\mathrm{KL}}(S\|T)$, from student S to teacher T. Which mismatch does its mode-seeking behavior strongly discourage?

   A. Substantial student mass where teacher probability is near zero.

   B. Student mass being split evenly between two equally likely teacher modes.

   C. Student and teacher assigning identical probabilities to likely tokens.

   D. A student missing a teacher mode to which the student assigns almost no mass.

7. (2 points) GRPO samples two responses to each of two teddy bear story prompts. Rewards are $(8, 10)$ for the first prompt and $(2, 4)$ for the second. Advantages are normalized within each prompt's group. What are the signs of the advantages for the responses with rewards 8 and 4?

   A. Reward 8: positive. Reward 4: negative.

   B. Reward 8: positive. Reward 4: positive.

   C. Reward 8: negative. Reward 4: negative.

   D. Reward 8: negative. Reward 4: positive.

8. (2 points) A policy writing teddy bear stories gains reward by adding irrelevant praise that its learned reward model favors. Human ratings of story quality decline. Which intervention directly addresses the faulty reward signal?

   A. Increase the rollout batch size to estimate the same reward more precisely.

   B. Revise reward training data to separate story quality from praise.

   C. Train the value model longer to predict the praise-related reward better.

   D. Increase the PPO clipping range to allow faster gains in the same reward.

9. (3+4 points) **What PPO clipping does.** For one sampled action, let $\rho = \pi_\theta(a \mid s)/\pi_{\text{old}}(a \mid s)$. With A fixed and all other terms ignored, PPO maximizes

$$J(\rho, A) = \min(\rho A, \text{clip}(\rho, 0.8, 1.2)A)\,.$$

(i) At $\rho = 1.4$, give the clipped ratio and compute J for $A = +2$ and $A = -2$. Show which term the minimum selects in each case.

(ii) For each case at $\rho = 1.4$, explain whether maximizing J locally favors increasing ρ, decreasing it, or neither. Does clipping enforce $0.8 \leq \rho \leq 1.2$ on the actual policy? Justify.

(i) - **Clipped ratio (1 point):** $\text{clip}(1.4, 0.8, 1.2) = 1.2$.

- **Positive advantage (1 point):** For $A = +2$, $J = \min(2.8, 2.4) = 2.4$.

- **Negative advantage (1 point):** For $A = -2$, $J = \min(-2.8, -2.4) = -2.8$.

(ii) - **Positive advantage (1 point):** Flat objective, so no incentive to increase ρ.

- **Negative advantage (1 point):** $J = -2\rho$, so ascent favors decreasing ρ.

- **No hard bound (1 point):** No. Clipping does not enforce $0.8 \leq \rho \leq 1.2$ on the actual policy.

- **Surrogate only (1 point):** Clipping only limits the surrogate incentive. Shared parameters and other samples can still move actual ratios outside $[0.8, 1.2]$.

10. (4+2 points) **Learning from outcome rewards.** Four responses to the same teddy bear counting problem receive final-answer rewards $(1, 1, 0, 0)$. Response 1 has valid reasoning. Response 2 has a false intermediate claim but the correct final answer. GRPO assigns every token in response i the advantage

$$A_i = \frac{R_i - \bar{R}}{s_R}, \qquad s_R = \sqrt{\frac{1}{4}\sum_{j=1}^{4}(R_j - \bar{R})^2}.$$

(i) Compute $\bar{R}$, s_R, and all four advantages. Explain whether the advantage assigned to each token can distinguish the false claim in response 2 from valid steps in response 1.

(ii) Describe how on-policy distillation can supply a token-level signal: specify which model generates the prefixes and what feedback is used. Give one reason it may still miss reasoning errors.

(i) - **Statistics (1 point):** $\bar{R} = 1/2$ and $s_R = 1/2$.

- **Advantages (1 point):** $A = (1, 1, -1, -1)$.

- **Credit assignment (1 point):** No. This advantage cannot locate the false claim in response 2.

- **Same advantage (1 point):** Both successful responses get $+1$ at every token, including the false claim.

(ii) - **Local signal (1 point):** Match teacher token probabilities on student rollout prefixes.

- **Limitation (1 point):** Teacher errors, or plausible continuations of a false prefix.

*
* *

We hope you enjoyed this exam and the class so far. Looking forward to spending the rest of the quarter together!